



Quiz I

Algorithms Analysis & Design

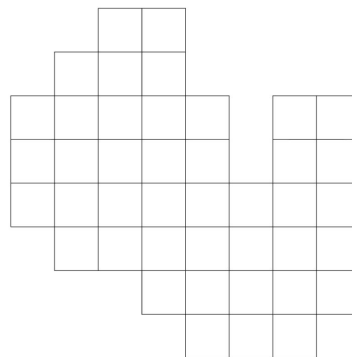
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Directions



1. Keep all belongings near the board.
2. Don't use the seats at the back.
3. Fill in Details in the answer sheet.
4. NO additional sheet will be given even for rough work.
5. Any malpractice will result in 0 marks.

Questions



- Can the following figure be tiled by dominos? A domino being 2 adjacent (horizontally or vertically) squares. Give a tiling or a short proof that no tiling exists.

- Give proof of correctness and runtime analysis for the PeasantMultiply algorithm for multiplying 2 numbers.

PEASANTMULTIPLY(x, y):

```

prod ← 0
while x > 0
    if x is odd
        prod ← prod + y
    x ← ⌊x/2⌋
    y ← y + y
return prod
    
```

x	y	$prod$
		0
123	+ 456 =	456
61	+ 912 =	1368
30	1824	
15	+ 3648 =	5016
7	+ 7296 =	12312
3	+ 14592 =	26904
1	+ 29184 =	56088

- Show that if v is a column vector of length $n = 2^k$, then the matrix-vector product $H_k v$ can be calculated using $O(n \log n)$ operations.

The *Hadamard matrices* $H_0, H_1, H_2, \dots$ are defined as follows:

- H_0 is the 1×1 matrix $[1]$
- For $k > 0$, H_k is the $2^k \times 2^k$ matrix

$$H_k = \left[\begin{array}{c|c} H_{k-1} & H_{k-1} \\ \hline H_{k-1} & -H_{k-1} \end{array} \right]$$

